nature portfolio

Peer Review File

A foundation model for generalizable cancer diagnosis and survival prediction from histopathological images

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This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
In this manuscript, the authors developed the BEiT-based model Pre-training on Histopathological images (BEPH) for pathology image analyses. They compared the performance of BEPH with AlexNet, ResNet, VGG19, Efficient Net, DINO, and other machine learning methods. They reported improved performance of BEPH in cancer type classification and survival outcome prediction compared with these baseline methods. Below are my comments.

1. The authors downsampled all images to 224 x 224 pixels, which led to a substantial loss of information. Although this is a common practice in many early studies that used ImageNet-based models, it would be interesting to see how the choice of image patch size affects model performance and robustness.

2. Although BEPH showed good average performance in the binary cancer versus benign tissue classification task, the variance of patient-level and image-level accuracy is substantial in many models under comparison. It would be helpful if the authors conduct statistical analyses to examine the significance of the observed performance difference given the large variance.

3. The authors found that different levels of magnification resulted in different diagnostic performance. A few recent studies found similar differences and proposed methods to aggregate predictions from different levels of magnification. The authors could employ related ensemble methods to aggregate prediction results from various levels of magnification.

4. There are a few recently released foundation models, including Prov-GigaPath, cTransPath, and CHIEF. The authors could compare their results with these newer methods that involved substantial model pertaining.

5. The presented analyses of data point clusters are mostly qualitative. For example, although the data point clusters appeared to be tighter when the training data size increased, no quantitative justification was provided to support this observation. It would be helpful if the authors employ quantitative measurements to strengthen their claims on the clustering patterns.

6. The results related to survival outcome prediction have a wide range of c-indices. In the comparison between different machine learning models, the ranges of c-indices overlap substantially. The added value of BEPH in this study setting appears less clear.

7. Related to the previous point, BEPH seem to have worse performance than a few conventional methods (e.g., ABMIL) in some disease types. It would be interesting to investigate the potential reasons underlying these observations. This could provide readers with additional insights regarding the most suitable scenarios to apply these machine learning model with different characteristics.

8. There are a few cancer-specific studies on the prediction of survival outcomes using histopathology images. Many of them have better performance than off-the-shelve machine learning models for pathology imaging analyses. Below are a few examples. It would be interesting to compare the performance of these studies with the results presented in the current manuscript.
https://pubmed.ncbi.nlm.nih.gov/29467373/

https://pubmed.ncbi.nlm.nih.gov/32895914/
https://pubmed.ncbi.nlm.nih.gov/33722896/
https://arxiv.org/abs/1902.03582

9. The authors visualized the feature space of models pre-trained on ImageNet and histopathology images, those pre-trained on ImageNet only, and models derived from randomly initialized parameters. Although some patterns were observed, the visualization and interpretation were largely subjective. The authors could further provide measurements on the vicinity of data points belonging to different categories in various study settings to better justify their interpretations.

10. The attention heatmap visualization is restricted to a limited number of selected images. It is unknown if the highlighted observation is generalizable to all cases included in this study. The authors could quantify the amount of attention associated with each tissue region across all samples, which will ensure that the identified observation is generally applicable to most cases.

11. Currently, the only ablation study presented in this analysis compares BEPH with a baseline model that is only pre-trained with ImageNet data. This baseline for comparison is understandably a low baseline due to its limited comprehension of pathology patterns. A more interesting ablation study might be to investigate the contribution of the mask image modeling (MIM) component in their current model design.

(Remarks on code availability)
The GitHub repository provides a README file with sufficient instructions for running the application.

Reviewer #2

(Remarks to the Author)
In this paper, the authors propose an artificial neural network specifically designed for digital pathology, which was pretrained on ImageNet-1k and fine-tuned on patches extracted from The Cancer Genome Atlas . One of the advantages of the model is its size of 86 Mparameters. The another advantage, is the method evaluation through several scenarios, including cancer patch-level recognition tasks, cancer WSI-level classification tasks, and survival prediction tasks for cancer subtypes. However, selected dataset foa a patch level scenario came foram a soingle institution, so the method robustness was not investigated. The main disadvantage is the lack of comparison to state-of-the-art results, such as the UNI model [1] or the Giga-Path model [2], which are recognized foundational models in digital pathology. Additionally, the paper's style is difficult to follow, and the sections are poorly organized. Without such comparisons, the novelty of the method remains unverified.

Specific Comments:

1. Lack of Details about Patch Size: The authors state, “resulting in a pre-training patch dataset of 11,774,353 samples.” However, there are no details regarding the patch size utilized.

2. Formatting Issues: There are several formatting inconsistencies, such as “BEPH-CLAMSurvival a” and “acquisition18.”

3. Comparison with Foundation Models: The paper lacks comparison with existing foundational models for digital pathology, such as UNI or GigaPath. The authors state that "Compared to the state-of-the-art models, including traditional transfer learning models, weakly supervised models, and contrastive learning models (DINO), BEPH consistently achieves superior performance and label efficiency." This claim requires substantiation.

4. Evaluation of Image Detail Loss: The statement “resulting in a loss of a significant amount of image details” lacks clarity. The authors should provide a basis for this assessment, such as statistical analysis or tests conducted to confirm the loss of detail.

4. References to CNN Models: The statement referring to “the latest reported CNN models and weakly supervised models” must specify which CNN models are being referenced. A lack of literature citations makes this point vague.

5. Figure Formatting: In Figure 2.f, the size of the caption is excessively large, which may distract from the content. In Figure 2.e, providing additional details about model size would be beneficial.

6.Claim Regarding Clinical Potential: The assertion that “MIM-based pre-training enhances the adaptability of the model to pathological image resolution and cancer type transformation” lacks validation. The clinical potentialis not demonstrated, particularly given that the dataset used in this study is relatively small and derived from a single institution.

7. Section Organization: The overall organization of the sections requires improvement for better clarity and flow.

8. Misidentification of ResNet Models: The claim that "the BEPH outperforms the ResNet supervised pre-trained" lacks specificity; which ResNet model is referenced here?

9. Consistency in Formatting: There is inconsistency in formatting, such as “11,774,353 samples” versus “1.28 million.”

10. Comparison with Existing Datasets: The authors state, “Firstly, although our self-supervised pre-training dataset is much larger compared to currently reported datasets.” This is misleading as referenced in the paper by Xu et al. [2], which reports “Prov-GigaPath, a whole-slide pathology foundation model pretrained on 1.3 billion 256 × 256 pathology image tiles in 171,189 whole slides from Providence, a large US health network comprising 28 cancer centers."

11. Training Time Clarification: The authors mention that "Finally, our model is pretrained for 340 hours" without stating the stop-training criteria or the resources utilized.

12. Lack of Segmentation Details: The statement “SI begins with the automatic segmentation of the tissue regions” lacks detailed methodology. More information is needed on how this segmentation was achieved.

13. Robustness of Patch-Level Classification Task: The dataset utilized for patch-level classification contains samples from a single institution; therefore, the robustness of the method is neither investigated nor demonstrated.

14. Dataset Links: The dataset section should provide links to each dataset utilized in the study.

15. Inconsistent Terminology: Phrases such as “Our public BRCA,” “Our public NSCLC,” and “Our public RCC” are inappropriate since these datasets were not provided by the authors.

16. Careless Presentation: Overall, the text appears to be prepared in a careless manner, requiring substantial revision to enhance clarity and professionalism.

[1] Chen, R. J., Ding, T., Lu, M. Y. et al. Towards a general-purpose foundation model for computational pathology. Nat Med 30, 850–862 (2024). https://doi.org/10.1038/s41591-024-02857-3
[2] Xu, H., Usuyama, N., Bagga, J. et al. A whole-slide foundation model for digital pathology from real-world data. Nature 630, 181–188 (2024). https://doi.org/10.1038/s41586-024-07441-w

(Remarks on code availability)

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
The authors have addressed the comments raised previously. Thank you.

(Remarks on code availability)
The code provides a README file with enough instructions for installing and running the application.

Reviewer #2

(Remarks to the Author)
The authors addressed most of the comments; however, not all explanations are accurate.

For example, "For the survival prediction task, BEPH achieves the best performance across most cancer types, except for CCRCC, where UNI demonstrates superior results. This highlights BEPH's strong performance in survival prediction across various cancer types." - However, based on the included table, this statement is not entirely accurate. BEPH outperforms UNI significantly in only two cases: CRC and PRCC. Additionally, there is a lack of statistical analysis, which means we cannot determine whether the observed differences are statistically significant.

(Remarks on code availability)
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Dear Reviewers:

Thank you for the reviewers' comments concerning our manuscript entitled " A foundation model for generalizable cancer diagnosis and survival prediction from histopathological images "(ID: NCOMMS-24-32882). Those comments are all valuable and very helpful for revising and improving our paper, as well as the important guiding significance to our research. We have carefully addressed your comments and made corrections. The detailed responses and the corresponding changes in the manuscript are shown in the following pages. Representative results have been highlighted in our response, and comprehensive tables and figures presenting all results are attached.

We hope these revisions align with your expectations and improve the overall quality of the manuscript.

Response to Reviewer #1:

1. The authors downsampled all images to 224 x 224 pixels, which led to a substantial loss of information. Although this is a common practice in many early studies that used ImageNet-based models, it would be interesting to see how the choice of image patch size affects model performance and robustness.

Thank you for your insightful comment regarding the choice of image patch size. To address the potential impact of downsampling images to 224×224 pixels on model performance and robustness, we conducted experiments on the LC25000 and BreakHis datasets, using multiple image sizes: the original resolution (original size × original size), 512×512, 384×384, and 224×224. Our model architecture comprises a pretrained ViT-base model with a single linear classifier layer.

Before presenting the results, we will reiterate the calculation method of accuracy for the BreakHis dataset follow the original paper[1].

patient level accuracy:

$$\text{Patient Score} = \frac{N_{rec}}{N_P}$$

Let N_p represents the number of pathological images owned by patient p. For each patient, N_{rec} is the number of pathological images that are correctly classified.

$$\text{patient level accuracy} = \frac{\sum \text{Patient Score}}{\text{Total Number of Patients}}$$

Image level accuracy:

$$\text{Image level accuracy} = \frac{N_{rec}}{N_{all}}$$

N_{all} represents the total number of pathological images from all patients, and N_{rec} represents the number of correctly classified pathological images among them.

In the Figure S1.a and S1.b (Page 4 in this reply , Supplementary Fig. 4, Page 4 in Supplementary), we present the average accuracy and AUC for both the LC25000 and BreakHis datasets across these patch sizes. The results indicate that accuracy and AUC remain stable above 90% for all patch sizes. Specifically, for the LC25000 dataset, accuracy hovers around 99% across different patch sizes, while for BreakHis, the image-level accuracy remains close to 93%, and the patient-level accuracy is approximately 94%. This is the first robustness evidence that the reduction in patch size does not substantially impact model performance.

Since the BreakHis dataset includes images at multiple resolutions (40X, 100X, 200X, and 400X), we further analyzed the impact of patch size across these magnifications (S1.c, S1.d, S1.e, S1.f). The figures illustrate image-level and patient-level accuracy and AUC at these different magnifications. The accuracy and AUC across all magnifications consistently exceeds 90%, showing only minimal fluctuations. For example, image-level accuracy across all patch sizes is around 94%, and patient-level accuracy reaches approximately 95% ,at the 40X resolutions. Even at larger patch sizes, such as 384×384 and 512×512, the accuracy and AUC only deviate by around 1–2%, reinforcing our model's robustness to variations in both resolution and patch size.

It is worth noting that the performance at the 224×224 patch size is slightly higher than other sizes. This is likely because our model was pretrained on 224×224 images, giving it an advantage when processing images at this size. For instance,in the breakhis dataset, the accuracy and AUC of 40x resolution of 224 × 224 patch size are slightly better than other sizes. However, even at larger patch sizes, the model achieves comparable performance, indicating that downsampling to 224×224 does not lead to a substantial loss of model performance.

Besides, based on Reviewer 2's comment #4 (Page 29 in this reply), we conducted a detailed evaluation of the extent of detail loss when downsampling images using two widely adopted metrics in Table S12 (Page 30 in this reply): Structural Similarity Index (SSIM) and Peak Signal-to-Noise Ratio (PSNR). They confirm that some detail is indeed lost during downsampling, supporting our model's robustness to detail loss.

In summary, these findings confirm that our model demonstrates strong robustness across different patch sizes and resolutions. The consistently high accuracy and AUC across all tested conditions highlight the model's reliability and performance. This robustness, combined with our model's high performance across both LC25000 and BreakHis datasets, justifies the use of 224×224 patch size in our experiments.

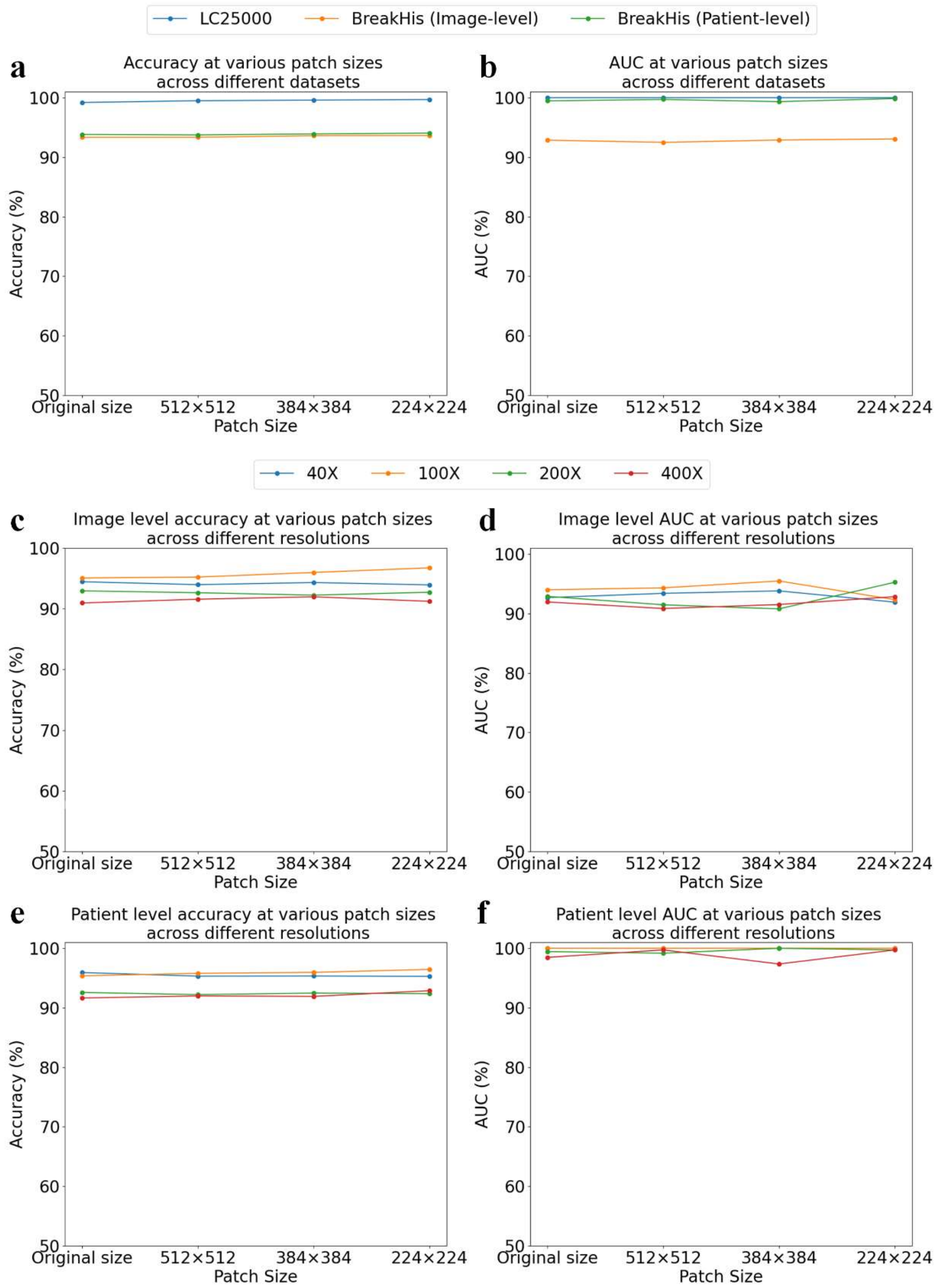

Figure S1 Model performance of BEPH on the LC25000 and BreakHis datasets across different patch sizes. a ,b. Accuracy and AUC for LC25000 and BreakHis across varying patch sizes, with BreakHis values averaged across resolutions. c, d, e, f. Accuracy and AUC at image and patient levels for BreakHis under different resolutions and patch sizes.

2. Although BEPH showed good average performance in the binary cancer versus benign tissue classification task, the variance of patient-level and image-level accuracy is substantial in many models under comparison. It would be helpful if the authors conduct statistical analyses to examine the significance of the observed performance difference given the large variance.

Thank you for your valuable comment on scalability. Following Matthew et al.[2], Zeng et al.[3], we conducted the independent two-sample t-test to assess the significance of performance differences between models at both patient and image levels. To ensure independence, we regenerated the training and testing datasets separately for each method. It is worth noting that, due to the incomplete implementation of many comparative models, reproduction of their results was not feasible. Therefore, we rely on the results reported in the original papers, assuming them to represent the upper performance bounds of these models to ensure a relatively fair comparison.

The results of the significance tests have been updated in Figure 2 (Page 6 in the revised manuscript, which corresponds to Figure S2 in this reply). Significance is indicated by "*" for p-values < 0.05 and non-significance indicated by "ns".

In patient-level accuracy (Figure S2), BEPH showed significantly higher accuracy than most models, including GLPB, Deep, MILCNN, SW, and RPDB. No significant differences were found between BEPH and MIL-NP or MPCS-RP. In image-level accuracy (Figure S2), similarly, BEPH outperformed other models, except for comparisons with MPCS-RP. MPCS-RP was not significantly different at both levels, but BEPH still had the highest average.

These results confirm BEPH's superior performance across several cases, addressing concerns about performance variance.

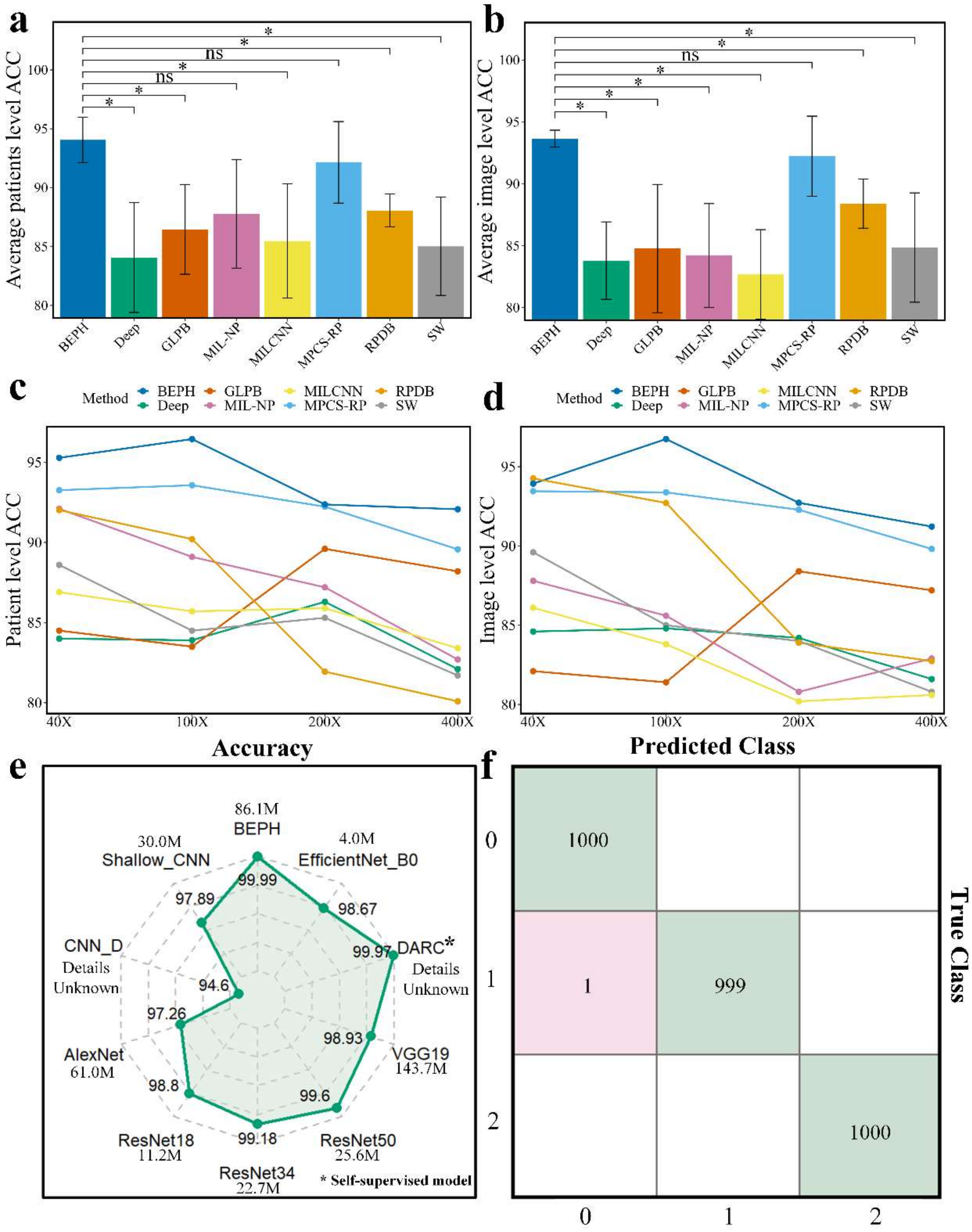

Figure S2 Performance evaluation on publicly available datasets composed of patches. a, Average accuracy comparison at the patient level for different resolutions. b, Average accuracy comparison at the image level for different resolutions. The compared models are the same as a). Training and testing sets for each method were independently split, with significance indicated by “*” and non-significance by “ns,” calculated using an independent two-sample t-test. c, d, Performance of different methods at patient level and image level, respectively, at magnification of 40, 100, 200, and 400. e, Performance of different algorithms on the LC25000 dataset. f, Confusion matrix on validation set of LC25000 dataset.

3. The authors found that different levels of magnification resulted in different diagnostic performance. A few recent studies found similar differences and proposed methods to aggregate predictions from different levels of magnification. The authors could employ related ensemble methods to aggregate prediction results from various levels of magnification.

Non-ensemble model	Ensemble model
ACC	ACC
0.8510	0.9873

Table S1 Results of ensemble voting under different magnification levels.

Thank you for your valuable suggestions. We indeed explored the impact of integrating different magnification levels on diagnostic performance using the BreakHis dataset. Since images at different magnification levels were collected from the same patient, the evaluation of ensemble effects could only be conducted at the patient level. However, as noted in Reviewer 1's comment #1, evaluating performance at the patient level differs from conventional approaches; therefore, we only reported the patient ACC results.

Our ensemble approach employs a simple voting-based strategy: the patient scores generated by classification models at different magnification levels are aggregated, and the highest patient score is used to calculate the final patient ACC. The results show a significant improvement in performance with the ensemble approach. Specifically, the average patient ACC for individual magnification levels is 94.05±1.93, while the average patient ACC using the voting ensemble method increased to 98.93±0.33. These findings indicate that the ensemble strategy proposed by the reviewer indeed has a positive impact.

However, we would like to emphasize that we chose not to adopt an ensemble approach in the primary analysis for the following reasons. First, the number of images collected under different magnification levels varies among patients. In some cases, the patient scores for certain magnification levels may be overestimated due to the limited number of images, which could inflate the patient ACC. Second, our primary objective is to develop a foundation model and evaluate its performance on this task. To maintain the method's simplicity within a reasonable scope, we intentionally avoided additional complexity, which is why we only added a classification layer for the downstream tasks.

4. There are a few recently released foundation models, including Prov-GigaPath, CTransPath, and CHIEF. The authors could compare their results with these newer methods that involved substantial model pertaining.

Thank you for comments. We agree that comparisons with the latest methods, including Prov-GigaPath, UNI, CTransPath, and CHIEF, are essential. These pathology foundation models have recently demonstrated outstanding performance on tasks such as WSI-level classification and survival prediction. Fortunately, the backbones of these models are also based on ViT, making it feasible to compare them within our downstream task framework by replacing the backbone with different pretrained ViT models.

The results, as shown in Table S2 (Supplementary Table 12 in supplementary), indicate that BEPH lags behind UNI and Prov-GigaPath in performance when using either 25% or 100% of the training data. This is reasonable, as these models were trained on significantly larger datasets compared to ours. Besides, GigaPath utilizes the ViT-G backbone, which is a deeper network with significantly more parameters. However, when using 100% of the training data, our model outperforms both CTransPath and the recently published CHIEF in classifying three cancer subtypes. But BEPH is worse at enduring data hunger than CHIEF.

For the survival prediction task, as shown in Table S3 (Supplementary Table 13 in supplementary), BEPH achieves better performance than other methods across six cancer types in most cases, with the exception of CCRCC, where UNI outperforms BEPH, and STAD, BRCA, where CHIEF performs better. These results across two WSI-level tasks demonstrate that BEPH offers unique advantages in certain scenarios compared to other state-of-the-art models.

Methods	BRCA Subtypes Classification		Lung Subtypes Classification		Kidney Subtypes Classification	
	25% train data	100% train data	25% train data	100% train data	25% train data	100% train data
BEPH	0.8547± 0.0433	0.946± 0.0188	0.9095± 0.0315	0.9719± 0.0051	0.9844± 0.0027	0.9941± 0.0013
CHIEF	0.9212± 0.0181	0.9446± 0.0111	0.9235± 0.0221	0.966± 0.0060	0.9897± 0.0053	0.9983± 0.0019
CtransPath	0.867± 0.0460	0.9458± 0.0122	0.9113± 0.0103	0.9615± 0.0088	0.9863± 0.0046	0.9989± 0.0012
UNI	**0.9733±0.0083**	**0.9827± 0.0056**	**0.9649± 0.0095**	**0.9783± 0.0093**	**0.9994± 0.0010**	**0.9996± 0.0008**
GigaPath	0.9365±0.0114	0.975± 0.0105	0.9508± 0.0165	0.9756± 0.0063	0.9989± 0.0018	0.9994± 0.0008

Table S2 Additional comparative experiments between BEPH and recent foundation models on the WSI-Classification task (AUC ± std).

	BRCA	CCRCC	CRC	PRCC	LUAD	STAD
BEPH	0.6643± 0.033	0.6685± 0.0386	**0.6758± 0.0721**	**0.7135± 0.1557**	**0.6039± 0.0762**	0.5941± 0.0478
CHIEF	**0.686± 0.0364**	0.6482± 0.0643	0.4844± 0.0635	0.6973± 0.1518	0.5533± 0.0941	**0.6372± 0.0391**
CtransPath	0.5653± 0.0473	0.6158± 0.0251	0.6027± 0.0658	0.559± 0.1432	0.5116± 0.0497	0.541± 0.0378
UNI	0.6519± 0.0652	**0.709± 0.037**	0.6007± 0.0866	0.6212± 0.0703	0.5713± 0.0909	0.6254± 0.0248
GigaPath	0.5642± 0.0279	0.6644± 0.0251	0.5737± 0.1266	0.6555± 0.1182	0.5623± 0.0521	0.5375± 0.0571

Table S3 Additional comparative experiments between BEPH and recent foundation models on the survival prediction task (C-index ± std)

5. The presented analyses of data point clusters are mostly qualitative. For example, although the data point clusters appeared to be tighter when the training data size increased, no quantitative justification was provided to support this observation. It would be helpful if the authors employ quantitative measurements to strengthen

their claims on the clustering patterns.

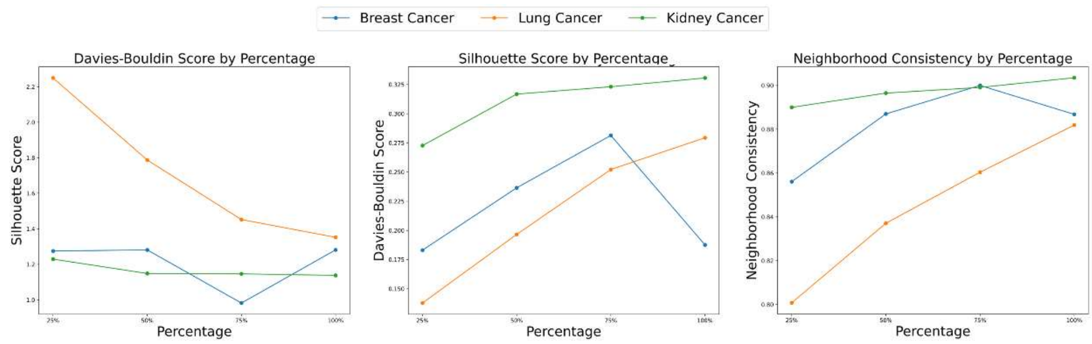

Figure S3 Quantitative assessment of feature space compactness under reduced training data. The clustering quality of cancer subtypes in feature space is evaluated using Davies-Bouldin Score[4,5], Silhouette Score[6,7], and Neighborhood Consistency[8] to quantify intra-cluster cohesion and inter-cluster separation. Different curve colors represent various cancer subtypes.

Thank you for your constructive comment. In response, we have conducted quantitative analyses to support our observation of clustering patterns with increasing training data size. For a quantitative assessment of the feature space, we formulated the task as a clustering evaluation problem without applying an explicit clustering algorithm. Clustering metrics were directly computed based on ground-truth labels, treating each label as a distinct "cluster". Using Davies-Bouldin Score[4,5], Silhouette Score[6,7], and Neighborhood Consistency[8] as quantitative metrics, we evaluated the clustering quality across different cancer types (breast, lung, and kidney cancer) while systematically increasing the training data proportion. The results, as shown in Figure S3 in this page (Supplementary Fig. 5, Page 5 in the Supplementary). The Davies-Bouldin Score generally decreases with larger training data, suggesting improved cluster separation. The Silhouette Score, which measures intra-cluster cohesion versus inter-cluster separation, shows an increase in most cases, further validating that data points are closer to their own clusters with higher data volumes. Finally, the Neighborhood Consistency improves across all cancer types as training data proportion grows, demonstrating greater clustering coherence. Interestingly, we observe a slight anomaly in the case of BRCA at 100% training data, where all metrics

(Davies-Bouldin Score, Silhouette Score, and Neighborhood Consistency) slightly decrease compared to 75%. This corresponds with a minor drop in the AUC for BRCA in the same setting, suggesting that, the addition of all training data may not only improve performance, but also introduce variability that slightly affects model performance. These quantitative metrics collectively affirm that increasing training data enhances clustering quality, while also highlighting specific cases where further analysis may be beneficial to understand the impact of data volume on model consistency.

Davies Bouldin Score:

Davies Bouldin Score is a metric used to evaluate clustering quality by measuring the trade-off between intra-cluster similarity and inter-cluster separation. A lower Davies-Bouldin Score indicates tighter clusters with greater separation between clusters, thus signifying better clustering performance.

Silhouette Score:

Silhouette Score is a metric that measures the cohesion and separation of clusters. It quantifies how well each data point fits within its assigned cluster relative to other clusters.The score ranges from -1 to 1, where values closer to 1 indicate better-defined clusters with clearer separation. Values near 0 suggest that the point lies between clusters, while negative values indicate that the point may have been assigned to the wrong cluster.

Neighborhood consistency:

For each data point, Neighborhood consistency uses the k-nearest neighbors algorithm to identify its k closest neighbors in the feature space and calculate the proportion of neighbors sharing the same class label, defining this proportion as the neighborhood consistency score. The values range from 0 to 1. A value of 0 indicates that all neighbors belong to different categories, representing complete inconsistency, while a value of 1 indicates all neighbors belong to the same category, representing complete consistency.

6. The results related to survival outcome prediction have a wide range of c-indices. In the comparison between different machine learning models, the ranges of c-indices overlap substantially. The added value of BEPH in this study setting appears

less clear.

Thank you for your valuable feedback. We acknowledge that survival outcome prediction results show substantial overlap in c-index ranges across different models, which may make BEPH's added value less immediately apparent. To address this concern, we provide the following clarifications and additional analyses.

Firstly, given the relatively small sample size (approximately 500 cases per cancer type), variability in survival times can impact c-index values, resulting in a wide range of scores across models.

Secondly, we used five-fold cross-validation due to the limited sample size. With only five folds, each validation set includes approximately 20% of the data, which may not fully represent the variability and heterogeneity in survival outcomes. This limited validation data can cause performance fluctuations across folds, particularly when small changes in data distribution significantly affect c-index values. However, increasing the number of folds to ten would further reduce each validation set's size, potentially compromising the reliability and stability of performance estimates, especially for deep learning models that require larger validation sets to accurately capture variance in predictions.

Despite this overlap, BEPH consistently demonstrates superior average c-index performance compared to other models (Table S4, Page 14 in this reply). Additionally, BEPH provides more stable and interpretable risk stratification in survival outcomes between high- and low-risk patient groups, as evidenced by Kaplan-Meier analysis with the lowest log-rank p-value. We believe this stability and interpretability represent the added value of BEPH (Figure S4, Page 4 in this reply).

Besides, to address the concern regarding the added value of BEPH and the overlap in c-indices with other models, we conducted paired t-tests and Cohen's d[9] to evaluate the model's performance differentiation across six cancer types in the survival prediction task.

When the p-value from the paired t-test is less than 0.05, it indicates a statistically significant difference in performance between the models. As shown in Table S5(Page 15, in this reply, Supplementary Table 5 in supplementary), BEPH demonstrates significant differences with most models on the BRCA dataset (except HIPT), while showing significant differences with a few models on the CRC and CCRCC datasets. However, on the LUAD, PRCC, and STAD datasets, BEPH does not exhibit statistically significant differences with other models.

Considering that the small sample size (only five-fold cross-validation) may limit the power of statistical tests, we referenced studies by Fan et al.[10], Tantithamthavorn et al.[11], and Hosseini et al.[12] and introduced Cohen's d to measure the effect size. Cohen’s d characterizes the effect size by relating the mean difference between models to the variability, analogous to a signal-to-noise ratio. According to conventional interpretations:

Small effect size when $0.2 <$ Cohen's d < 0.5,

Medium effect size when $0.5 \leq$ Cohen's d < 0.8,

Large effect size when Cohen's d ≥ 0.8.

Even with a small sample size, effect size can provide valuable information on the practical significance of differences between models. As seen in the results, only a few comparisons yield Cohen's d values below 0.5, indicating that the difference in c-indices between BEPH and other models has practical significance in most cases. This suggests that, despite some overlapping ranges in c-indices, BEPH offers meaningful differentiation in performance for survival outcome prediction.

Based on the suggestions, we have incorporated statistical analyses in the revised manuscript:

“*We conducted statistical tests using paired t-tests and Cohen’s d. Even when the paired t-test results were not significant, a Cohen’s d value greater than 0.5 indicates meaningful differences between methods. The test results are provided in Supplementary Table 5. Although the results of BEPH overlap with those of other methods, there are still observable differences.*” (Page 9,Line 257, **Result** section in the revised manuscript):

“*For the survival prediction task, we applied paired t-tests along with Cohen’s d to quantify effect sizes and performance differences across six cancer types, following methodologies from Fan et al.[54], Tantithamthavorn et al.[55], and Hosseini et al.[56].*” (Page 23,Line 680 **Statistical Analysis** section in the revised manuscript):

Method	BRCA	CCRCC	CRC	PRCC	LUAD	STAD
ABMIL	0.4871± 0.0706	0.5612± 0.0659	0.5656± 0.0669	0.6712± 0.0683	0.584± 0.0481	0.5618± 0.0439
DS-MIL	0.5338± 0.0539	0.591± 0.0834	0.5382± 0.044	0.636± 0.059	0.5918± 0.0626	0.5131± 0.062
GCN-MIL	0.4722± 0.0183	0.5484± 0.0511	0.4702± 0.0477	0.6539± 0.1195	0.5373± 0.0543	0.5459± 0.042
DeepAttnMISL	0.4716± 0.0207	0.5207± 0.0749	0.5613± 0.0786	0.4718± 0.1448	0.563± 0.0327	0.5633± 0.0596
HIPT	0.6344± 0.0449	0.6416± 0.0251	0.6081± 0.0784	0.6698± 0.0584	0.5377± 0.0397	0.5700± 0.072
CLAMSurvival (ResNet)	0.5946± 0.0484	0.6466± 0.0405	0.5502± 0.1187	0.599± 0.1368	0.5520± 0.1002	0.5119± 0.0882
CLAMSurvival (DINO)	0.5987± 0.0316	0.6555± 0.0807	0.6207± 0.0399	0.6974± 0.1105	0.5687± 0.0549	0.5840± 0.0428
CLAMSurvival (CHIEF)	**0.686± 0.0364**	0.6482± 0.0643	0.4844± 0.0635	0.6973± 0.1518	0.5533± 0.0941	**0.6372± 0.0391**
CLAMSurvival (CtransPath)	0.5653± 0.0473	0.6158± 0.0251	0.6027± 0.0658	0.559± 0.1432	0.5116± 0.0497	0.5410± 0.0378
CLAMSurvival (UNI)	0.6519± 0.0652	**0.709± 0.037**	0.6007± 0.0866	0.6212± 0.0703	0.5713± 0.0909	0.6254± 0.0248
CLAMSurvival (GigaPath)	0.5642± 0.0279	0.6644± 0.0251	0.5737± 0.1266	0.6555± 0.1182	0.5623± 0.0521	0.5375± 0.0571
CLAMSurvival (BEPH)	0.6643± 0.033	0.6685± 0.0386	**0.6758± 0.0721**	**0.7135± 0.1557**	**0.6039± 0.0762**	0.5941± 0.0478

Table S4 C-index (mean ± std) across six cancer types, derived from five-fold cross-validation. Bold indicates the best results.

	BRCA		CRC		CCRCC		LUAD		PRCC		STAD	
validation	p-value	Cohen's d	p-value	Cohen's d	p-value	Cohen's d	p-value	Cohen's d	p-value	Cohen's d	p-value	Cohen's d
BEPH --ABMIL	**0.0205**	**1.6637**	0.1292	**0.8527**	0.1001	**0.953**	0.3623	0.1747	0.6891	0.1925	0.2065	**0.6736**
BEPH --GCN-MIL	**0.0012**	**3.6999**	**0.0131**	**1.8736**	0.2538	**1.3216**	0.8438	**0.623**	0.3917	0.278	**0.047**	**0.6743**
BEPH --DS-MIL	**0.0033**	**2.7934**	**0.0138**	**1.9031**	**0.0417**	**0.5955**	0.236	0.094	0.5678	0.429	0.2061	**1.2693**
BEPH --HIPT	0.3074	**0.5227**	0.0922	**0.5525**	**0.0167**	0.4675	0.2523	**0.5572**	0.1338	0.1888	0.5946	0.3781
BEPH--DeepAttnMISL	**0.0002**	**5.6736**	0.2843	**0.9855**	0.3548	**1.7700**	0.2808	**0.5977**	0.6946	**0.8391**	0.4456	0.2582
BEPH—CLAMSurvival (Resnet)	**0.0137**	**1.8769**	0.0688	**1.1053**	**0.0245**	**1.572**	0.1183	**0.887**	0.3837	0.4371	0.1192	**0.8843**
BEPH --CLAM(DINO)	0.0732	**1.0796**	0.1225	**0.8734**	0.6299	0.233	0.0916	**0.9886**	0.8179	0.1099	0.6854	0.195
BEPH --CLAMSurvival (UNI)	0.7889	**0.128**	0.0529	**1.2171**	0.0356	1.393	0.0574	**1.1818**	0.1878	0.7096	0.3773	0.4436
BEPH --CLAMSurvival (CHIEF)	0.5438	**0.2963**	0.0202	**1.6718**	0.2573	0.5904	0.0916	**0.9884**	0.6151	0.2434	0.273	0.568
BEPH --CLAMSurvival (CtransPath)	0.0115	**1.9761**	0.1406	**0.8201**	0.0489	1.2513	0.0171	**1.7589**	0.0023	3.0928	0.1744	0.7376
BEPH --CLAMSurvival (GigaPath)	0.0007	**4.1754**	0.094	**0.9781**	0.8205	0.1084	0.1561	**0.779**	0.2286	0.6351	0.1022	0.9447

Table S5 Paired t-test and Cohen’s d effect size results. Bold indicates BEPH comparisons with other methods where p-value < 0.05 or Cohen’s d > 0.5, demonstrating significant differences between methods.

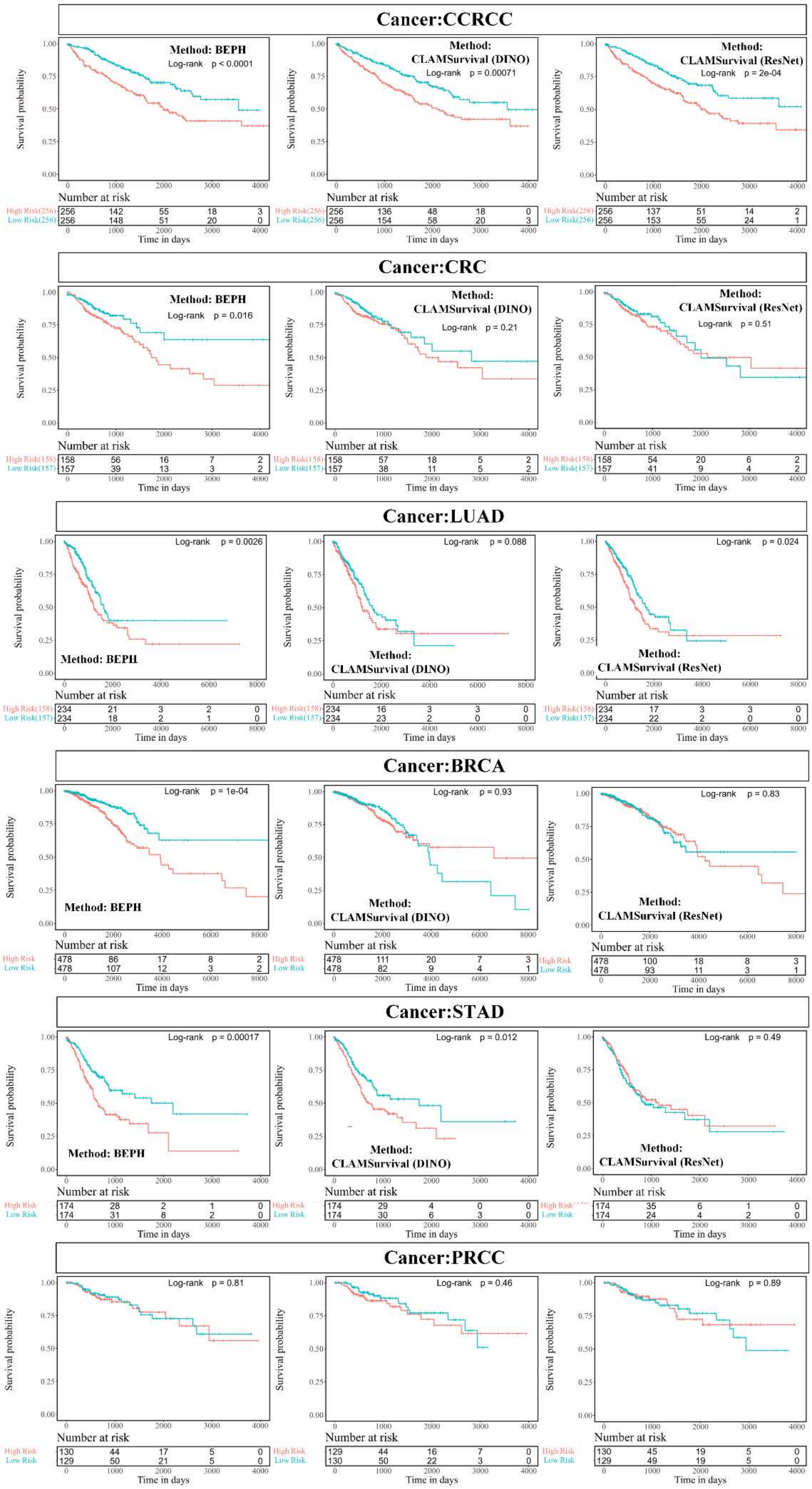

Figure S4 **Survival curves for patient stratification are depicted using Kaplan-Meier analysis.** To illustrate the added value of pathology image pretraining, the CLAMSurvival model was used with different initialization methods: BEPH, DINO (pretrained on pathology images), and ResNet (ResNet-50, pretrained on ImageNet-1K). Patients were stratified into high- and low-risk groups based on the output of the final CLAMSurvival layer, divided by the median. The number of patients who survived in each risk group is shown below the survival curves. The log-rank test calculates the p-value, indicating the significance of the difference in survival between the two groups. These curves illustrate the true survival rates of patients classified as high or low risk.

7. Related to the previous point, BEPH seem to have worse performance than a few conventional methods (e.g., ABMIL) in some disease types. It would be interesting to investigate the potential reasons underlying these observations. This could provide readers with additional insights regarding the most suitable scenarios to apply these machine learning model with different characteristics.

Thank you for your valuable comments. We appreciate your suggestion to examine the performance of BEPH relative to conventional methods. Upon further review, we confirm that BEPH achieves the highest C-index in all evaluated cancer types (as shown in our results table), demonstrating that its performance is not inferior to traditional methods like ABMIL(Table S4, Page 14 in this reply). Additionally, we would like to clarify that the horizontal lines in the box plots represent the median, not the mean. This may have led to some misunderstanding about BEPH's comparative performance. Given these results, we believe that BEPH's design consistently provides robust performance across different disease types, showcasing its adaptability and effectiveness in diverse scenarios.

Meanwhile, to clarify the boxplots Figure S5 in next page (Figure 4, Page 11 in the revised manuscript), we added an additional red dashed line to indicate the mean, and have updated both the figure and the corresponding text in the manuscript accordingly.

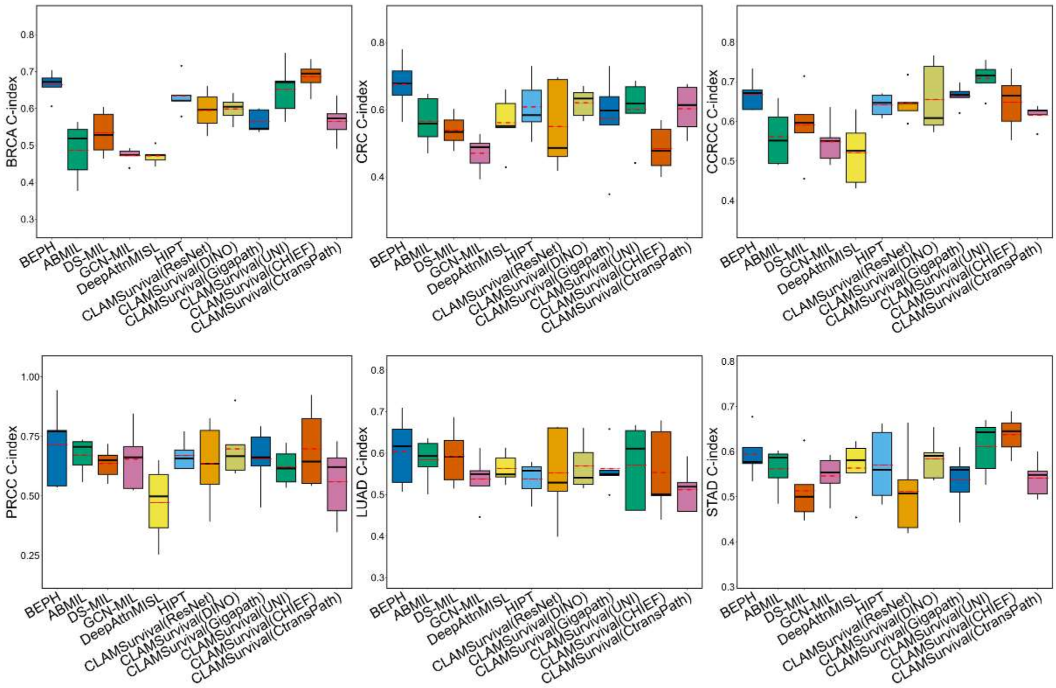

Figure S5 Different methods of C-index on six cancers. The boxplots displaying the 5-fold cross validation results. The black solid line indicates the median, and the red dashed line represents the mean.

8. There are a few cancer-specific studies on the prediction of survival outcomes using histopathology images. Many of them have better performance than off-the-shelve machine learning models for pathology imaging analyses. Below are a few examples. It would be interesting to compare the performance of these studies with the results presented in the current manuscript.

Thank you for your insightful comments. We acknowledge the importance of comparing our results with those of other cancer-specific studies mentioned in your feedback.

In the first, I would like to clarify that in the field of deep survival prediction, survival analysis methods can broadly be divided into two categories: regression and classification, each with distinct working principles and characteristics. Regression methods (such as the Cox proportional hazards model) are designed to directly model the relationship between event time and predictor variables, providing continuous risk predictions or survival time estimates. These methods are known for their interpretability, offering detailed insights such as hazard ratios for individual variables, which are useful for analyzing the specific effects of predictors on survival time. Classification methods, on the other hand, approach survival analysis as a classification task, stratifying patients into high- and low-risk groups with a focus on risk stratification rather than precise time prediction.

Therefore, our approach differs significantly from the referenced you provided. The cited studies on colorectal cancer, clear cell renal cell carcinoma, and renal cell carcinoma typically utilize slightly varied survival prediction methods but follow a similar approach. Each study transforms patient survival times into discrete labels, leveraging machine learning models (e.g., LASSO, logistic regression) or deep learning models (e.g., CNN, LSTM) for training and prediction. These methods frame survival prediction as a multi-class problem, applying cross-entropy as the loss function and using evaluation metrics such as AUC and ACC. Kaplan-Meier (KM) curves are then optionally used to stratify patients into high- and low-risk groups, enabling assessment of the model's prognostic accuracy. Notably, these studies have not made their code publicly available, and even if accessible, it would likely not be applicable for direct use. In contrast, our approach is based on an end-to-end Cox proportional hazards model using a neural network, which directly models survival outcomes and evaluates performance using the C-index (Concordance Index). This metric estimates the probability that the predicted survival times are in concordance with the observed outcomes, making it unsuitable for direct comparison with classification-based models evaluated using AUC or accuracy metrics.

To address your concern, we reviewed studies with similar methods (Table S6 in next page). Liu et al.[13] used the TCGA-BRCA dataset with multi-resolution image data and reported a C-index of 0.638, while our model achieved 0.6685 ± 0.0386. Wei et al.[14] proposed the MultiDeepCox-SC model for gastric cancer, achieving a C-index of 0.667, while our model's was 0.6643 ± 0.033, showing comparable performance. They did not provide all data or pre-trained weights, limiting direct comparison.

Additionally, Jiang et al.[15] developed a colorectal cancer survival model based on RetCCL, which achieved 0.5214 ± 0.0559 on our data, with our results being significantly better.

Study	Cancer Type	Reference C-index	Our C-index
Liu et al	TCGA-BRCA	0.638±0.0548	**0.6685 ± 0.0386**
Wei et al	TCGA-STAD	0.667	0.6643±0.0330
Jiang et al	TCGA-CRC	0.5214±0.0559	**0.6758±0.0721**

Table S6 Comparison of C-index between BEPH and the cancer-specific studies

Therefore, we believe that in the survival prediction task, BEPH demonstrates competitive performance across multiple cancer types compared to cancer-specific studies. Since BEPH is not limited to a particular type of cancer, it offers a greater advantage in terms of generalizability.

9. The authors visualized the feature space of models pre-trained on ImageNet and histopathology images, those pre-trained on ImageNet only, and models derived from randomly initialized parameters. Although some patterns were observed, the visualization and interpretation were largely subjective. The authors could further provide measurements on the vicinity of data points belonging to different categories in various study settings to better justify their interpretations.

Method	BEPH	BEPH without pretraining on histopathological images	BEPH without pretraining
Davies Bouldin Score	**2.4447**	2.5416	3.4517
Silhouette Score	**0.1624**	0.1372	0.0051
Neighborhood Consistency	0.9600	**0.9700**	0.7400

Table S7 Quantitative evaluation of model feature extraction capability based on NCT-CRC-HE-100K dataset

Thank you for your helpful suggestion. Similar to Reviewer 1's comment #5 (Page 9 in this reply), this question also concerns the quantitative measurement of the clustering and separation of different classes within the feature space. Similarly, we calculated clustering metrics using the Davies-Bouldin Score, Silhouette Score, and Neighborhood Consistency, directly based on ground-truth labels. As our result in Table S7 (Page 20 in this reply). (Supplementary Table 6, Page 10 in supplementary), the Davies-Bouldin Score was the lowest, the Silhouette Score the highest. Although the Neighborhood Consistency was not the highest, it was close to the best result. These metrics demonstrate improved class separation and intra-class compactness for the model pre-trained on histopathology images, supporting the observed patterns and providing a more objective interpretation of the feature space.

10. The attention heatmap visualization is restricted to a limited number of selected images. It is unknown if the highlighted observation is generalizable to all cases included in this study. The authors could quantify the amount of attention associated with each tissue region across all samples, which will ensure that the identified observation is generally applicable to most cases.

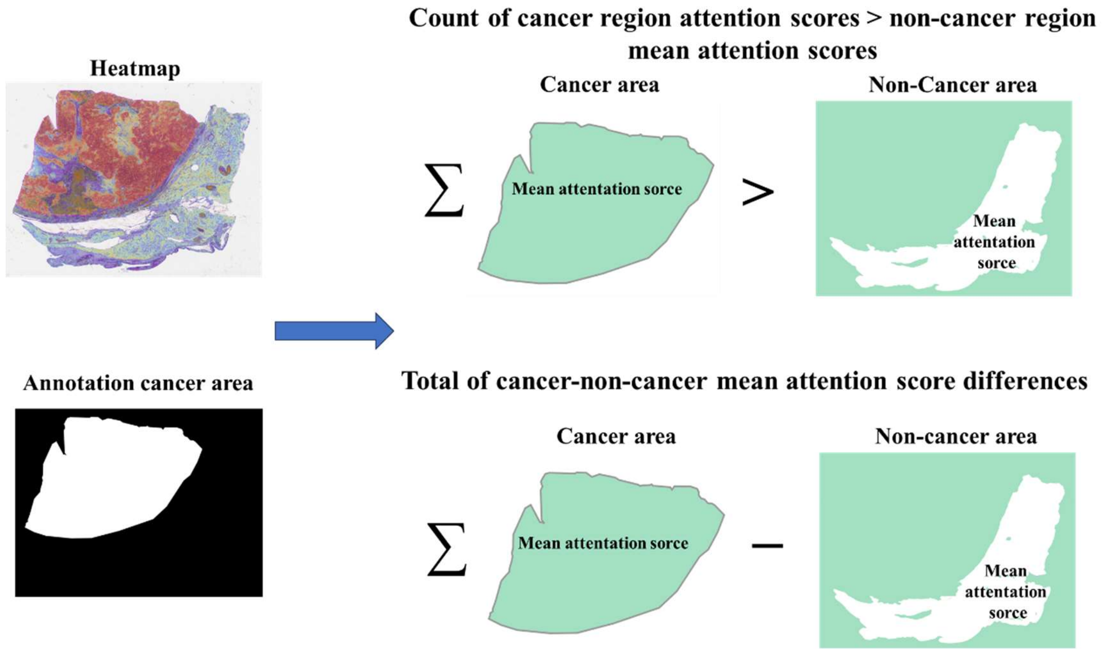

Figure S6 Quantitative analysis of model attention on Cancer Across All Cases.

While attention heatmaps generated by weakly supervised CLAM models are not designed for pixel-level annotation of regions of interest (ROIs), we conducted a quantitative evaluation to assess the feasibility of using these heatmaps as auxiliary annotation tools in clinical and research settings, and to verify the performance of the model's attention. For this purpose, we downloaded publicly available cancer region annotations from https://zenodo.org/records/5320076 , where annotations were performed by trained observers using QuPath v0.1.2 and later converted to CSV format. Using these annotations, we segmented each slide into cancerous and non-cancerous regions, obtaining a 0-1 attention score for each 224×224 patch within each region type. The mean attention scores were then calculated for both cancerous and non-cancerous regions. Finally, we quantitatively evaluated the effectiveness of different pre-trained backbones by comparing the cumulative mean attention scores in cancerous regions to those in non-cancerous regions, and by counting the number of images where the mean attention score for cancerous regions exceeded that for non-cancerous regions (shown in the Figure S6, Supplementary Fig. 6 in supplementary).

	Count of cancer region attention scores > non-cancer region mean attention scores			Total of cancer-non-cancer mean attention score differences			t test (p-value)			WSI Number		
	BRCA	NSCLC	RCC	BRCA	NSCLC	RCC	BRCA	NSCLC	RCC	BRCA	NSCLC	RCC
CLAM_SB pre-training with ResNet-50	41	793	458	-88.37	55.27	27.65	<<0.01	<<0.01	<0.01	866	860	793
CLAM_SB pre-training with DINO	**836**	796	721	144.92	94.76	172.02	<<0.01	<<0.01	<<0.01			
BEPH	835	**810**	**743**	**189.76**	**125.41**	**185.76**	<<0.01	<<0.01	<<0.01			

Table S8 Quantitative evaluation of attention heatmaps. p-value << 0.01 indicates p-value $< 10^{-10}$. p-value < 0.01 indicates $10^{-10} <$ p-value < 0.01.

As shown in Table S8 (Supplementary Table 7 in supplementary)., BEPH almost achieved the highest count and total score compared to other pretraining methods, with statistically significant differences in attention scores for cancer vs. non-cancer regions (p-value << 0.01). This confirms that BEPH focuses more strongly on cancer regions in most cases. Additionally, to address the limited initial examples, we have included attention maps for TCGA-NSCLC and TCGA-RCC in Figures S7 and S7 (Supplementary Fig. 7 and Supplementary Fig. 8 in supplementary).

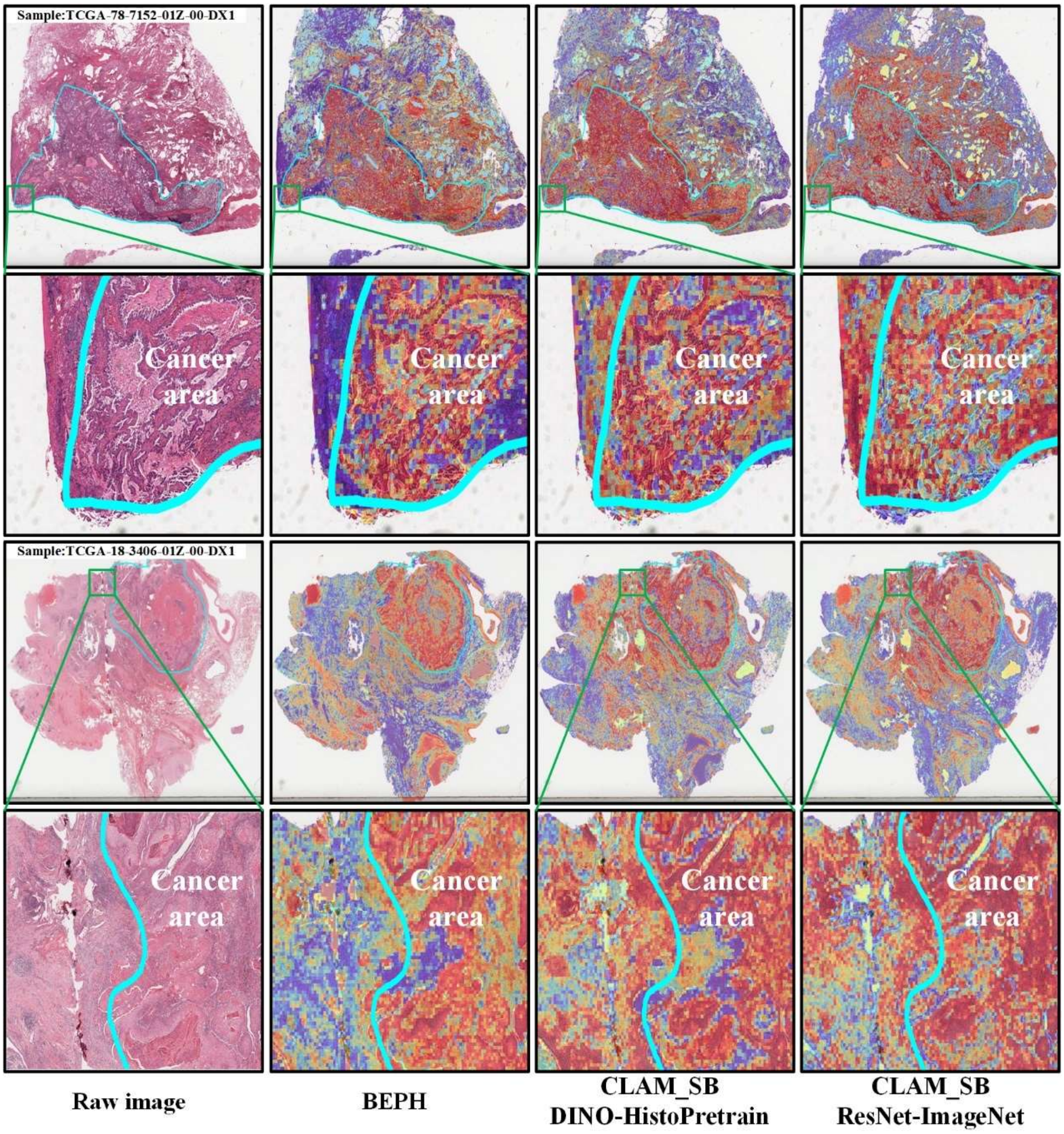

Figure S7 The heatmap of attention scores highlights significant morphological features of non-small cell lung cancer relevant for diagnosis. By visualizing attention scores in a spatial context, regions with higher scores indicate morphological features that the model considers relevant for distinguishing cancerous tissue from normal regions.

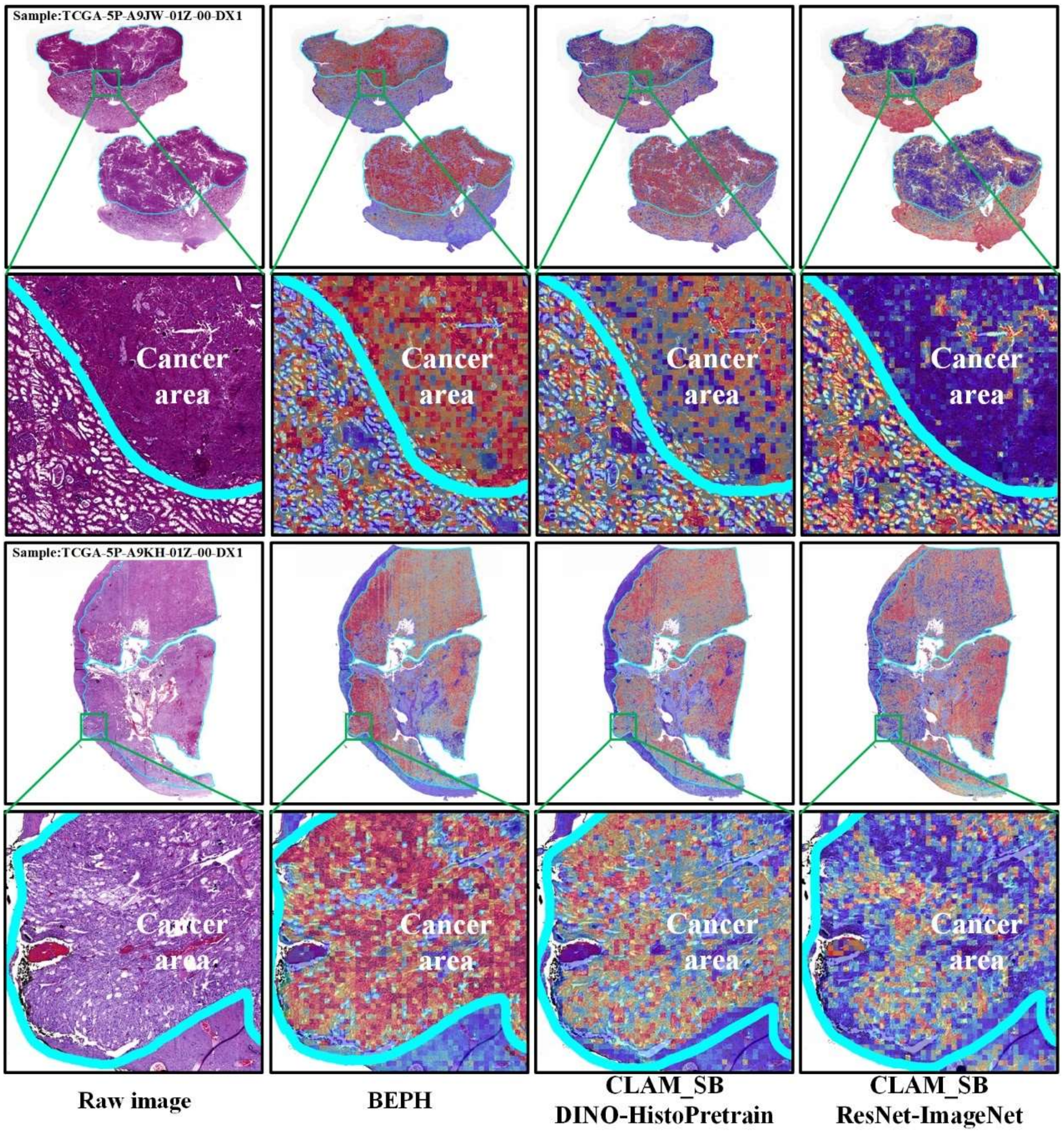

Figure S8 Similar to the Figure S7, the heatmap of attention scores highlights significant morphological features of renal cell carcinoma relevant for diagnosis. This visualization aids in interpreting the model's focus areas, enhancing the transparency of automated RCC diagnostic predictions.

11. Currently, the only ablation study presented in this analysis compares BEPH with a baseline model that is only pre-trained with ImageNet data. This baseline for comparison is understandably a low baseline due to its limited comprehension of pathology patterns. A more interesting ablation study might be to investigate the contribution of the mask image modeling (MIM) component in their current model

design.

Thank you for your comments. We conducted additional ablation experiments to determine the contribution of masked image modeling (MIM). In the original manuscript, our ablation study only compared the model performance differences between pathology-specific pretraining with a 40% mask ratio and directly using ImageNet pretraining. However, we did not investigate the impact of varying mask ratios on pathology-specific pretraining.

As is well-known, when the mask ratio is excessively high, the visible information in the input image becomes insufficient, making it challenging for the model to predict the masked regions based on the remaining context. This can lead to overly difficult pretraining tasks, causing the model to fail to converge effectively.

Due to the limited availability of NAVIA A100 computational resources at our institution and the high computational and economic costs of pretraining, we added an additional experiment with a reduced mask ratio of 10% to further evaluate the contribution of MIM(The mask ratio cannot be reduced to 0 because MIM pretrains by predicting the masked visual tokens.). Other pretraining parameters, such as training duration and learning rate, were kept consistent. After the same pretraining duration, the backbone was applied to three downstream tasks.

The results, presented in Table S9, Table S10, Table S11(Supplementary Table 8, Supplementary Table 9, Supplementary Table 10 in supplementary), show that a mask ratio of 40% achieves the best performance across all three tasks. In contrast, when the mask ratio was reduced to 10%, the performance significantly declined. This is because the MIM task primarily learns image features by predicting the masked visual tokens. When the mask ratio is too low, the pretraining task becomes too simple, as the model can easily reconstruct the masked regions based on the abundant visible information. As a result, the learned features are less meaningful, and the pretraining effectiveness is the poorest, even worse than directly using BEiTv2-ImageNet (both in accuracy and stability).

Method	image level				patient level				LC25000
	40X	100X	200X	400X	40X	100X	200X	400X	ACC
BEiTv2-ImageNet	**94.54±1.30**	95.03±0.44	92.36±0.95	89.62±5.58	95.59±0.59	94.79±0.90	92.48±4.06	90.32±0.47	99.88±0.19
BEPH（mask 10%）	91.97±0.90	92.79±0.69	90.55±0.72	89.40±1.08	93.49±2.20	94.04±0.77	91.57±0.78	90.68±1.10	97.96±2.42
BEPH（mask 40%）	93.93±0.31	**96.74±1.17**	**92.72±0. 7**	**91.22±1.19**	95.28±0.59	96.45±1.35	92.37±0.61	92.07±3.00	99.99±0.03

Table S9 Ablation experiment in patch-level classification (ACC percentage ± std).

	BRCA Subtyping		NSCLC Subtyping		RCC Subtyping	
	25% Training	100% Training	25% Training	100% Training	25% Training	100% Training
BEiTv2-ImageNet	0.8367±0.0653	0.9408±0.0159	0.8788±0.0218	0.9270±0.0131	0.9802±0.0097	0.9907±0.0062
BEPH (mask 10%)	0.6877±0.064	0.8795±0.0334	0.7043±0.0337	0.8496±0.020	0.8607±0.0635	0.9557±0.0093
BEPH (mask 40%)	**0.8547±0.0432**	**0.9460±0.0188**	**0.9095±0.0332**	**0.9719±0.0050**	**0.9844±0.0090**	**0.9941±0.0013**

Table S10 Ablation experiment in WSI-level classification (AUC percentage ± std).

	BRCA	CCRCC	CRC	PRCC	LUAD	STAD
BEiTv2-ImageNet	0.6440±0.0354	0.6826±0.0569	0.6677±0.0685	0.6740±0.1306	0.5587±0.1024	0.5575±0.0584
BEPH（mask 10%）	0.5400±0.0439	0.6429±0.0225	0.6353±0.085	0.4746±0.0937	0.5949±0.0993	0.5749±0.0407
BEPH（mask 40%）	**0.6643±0.033**	**0.6685±0.0386**	**0.6758±0.0721**	**0.7135±0.1557**	**0.6039±0.0762**	**0.5941±0.0478**

Table S11 Ablation experiment in survival prediction (C-index percentage ± std).

The updated ablation results can be found on Page 14 ,Line 342 of the revised manuscript.

"*To quantitatively evaluate the effect of the improvement, we performed patch-level evaluation on BreakHis, WSI-level evaluation on NSCLC subtype and survival prediction for BRCA, respectively. Table 1 shows that the performance of the model pre-training on* *pathological images is better than that of the model pre-trained only on ImageNet-1K (BEiTv2-ImageNet). This highlights the difficulty of SSL using natural images for specialized domains. Additionally, we demonstrated the importance of MIM in pretraining by reducing the mask ratio. An excessively low mask ratio can decrease the difficulty of the pretraining task, thereby limiting the model's learning capacity. In some cases, this can result in performance that is even worse than directly using BEiTv2-ImageNet. Complete ablation test results are in Supplementary Tables (Supplementary Table 8, Supplementary Table 9, Supplementary Table 10 in supplementary).*

Table 1 Ablation experiments on the effectiveness of pre-training.

	BreakHis（Mean magnifications）		***WSI weakly Classification***		***Survival prediction***
	Patient Level Accuracy	*Image Level Accuracy*	*NSCLC 25%*	*NSCLC 100%*	*BRCA*
BEiTv2-ImageNet	*93.30±1.51*	*92.89±2.07*	*0.8788±0.021*	*0.9270±0.013*	*0.6440±0.0354*
BEPH (mask 10%)	*92.45±1.21*	*91.18±0.85*	*0.7043±0.0337*	*0.8496±0.020*	*0.5400±0.0439*
BEPH (mask 40%)	***94.05±0.9928***	***93.65±0.6730***	***0.9095±0.0332***	***0.9719 ±0.0054***	***0.6643±0.0330***

"

Response to Reviewer #2

1. Lack of Details about Patch Size: The authors state, "resulting in a pre-training patch dataset of 11,774,353 samples." However, there are no details regarding the patch size utilized.

Thank you for your comment. We apologize for the oversight. The patch size used in our pre-training dataset is 224 × 224 pixels. We have updated the manuscript to include this information. Specifically, we have revised the text on Page 3, Line 104, to state: "*We have collected a total of 11.76 thousand pathology images (with 92 exclusions due to indeterminate magnification), resulting in a pre-training patch dataset of* ***11.77 million 224×224 pixel patches***."

2. Formatting Issues: There are several formatting inconsistencies, such as "BEPH-CLAMSurvival a" and "acquisition18."

Thank you for highlighting the formatting inconsistencies. We have addressed these issues in the revised manuscript. To ensure clarity, we have removed specific naming that may have caused confusion in the original manuscript. For example, in the original manuscript, on Page 4, line 115, we stated: *"Finally, we considered survival prediction tasks in BRCA, CRC, CCRCC, PRCC, LUAD, and STAD, named BEPH-CLAMSurvival (Fig. 1d). The data of BEPH-CLAM and BEPH-CLAMSurvival are from TCGA."*

In the revised manuscript, on Page 4, line 116, we have revised this to:

"Finally, we considered survival prediction tasks in BRCA, CRC, CCRCC, PRCC, LUAD, and STAD using the structure. The data used for WSI classification and survival prediction are sourced from TCGA."

Besides, we have corrected these issues to ensure consistency in terms such as "acquisition18" was due to an export formatting issue. We have revised the manuscript on page 3, line 74: " SL involves pre-training on labeled data, which is challenging due to the large gap between the speed of data annotation and data **acquisition**[18]."

3. **Comparison with Foundation Models: The paper lacks comparison with existing foundational models for digital pathology, such as UNI or GigaPath. The authors state that "Compared to the state-of-the-art models, including traditional transfer learning models, weakly supervised models, and contrastive learning models (DINO), BEPH consistently achieves superior performance and label efficiency." This claim requires substantiation.**

Thank you for your comments. First, we would like to point out that at the time of our comparisons, UNI and GigaPath had not yet been officially published, so they were not included in our original analysis. Instead, the primary comparison was made against HIPT, which is a pathology foundation model pretrained using DINO.

Additionally, we note that in the original manuscript, our model was shown to outperform the DINO-based HIPT model, not DINO v2 (which is the method used by UNI and GigaPath). This distinction may have caused some confusion.

Nonetheless, we have included comparisons with the latest models (Table S2, Table S3, Page 9 in this reply).

In the WSI-level classification task, BEPH underperforms compared to UNI and Prov-GigaPath when using either 25% or 100% of the training data. This is expected, as these models were trained on significantly larger datasets than ours. Additionally, GigaPath utilizes the ViT-G backbone, which is a deeper network with more parameters, contributing to its superior performance.

For the survival prediction task, BEPH achieves the best performance across most cancer types, except for CCRCC, where UNI demonstrates superior results. This highlights BEPH's strong performance in survival prediction across various cancer types.

Considering this, we have revised the statements in the original manuscript. Specifically, we replaced "contrastive learning models (DINO)" with "DINO v1 contrastive learning models" to clarify this point.

4. **Evaluation of Image Detail Loss: The statement "resulting in a loss of a significant amount of image details" lacks clarity. The authors should provide a**

basis for this assessment, such as statistical analysis or tests conducted to confirm the loss of detail.

Thank you for your insightful comment. To address your concern regarding the evaluation of image detail loss, we conducted a quantitative analysis using two widely accepted image quality metrics: SSIM(Structural Similarity Index), PSNR (Peak Signal-to-Noise Ratio),following Fischer et al.[16] and Helin[17]. As shown in Table S12 in this page(Supplementary Table 11 in supplementary), on the LC25000 and BreakHis datasets, the PSNR scores between the original image and the downsampled image (224 × 224) were 37.2733 and 34.8790, and the SSIM scores were 0.9308 and 34.8790. These results indicate a moderate loss of image details during the resizing process, which supports our original statement regarding the loss of details. while image quality metrics indicate a decrease, the accuracy and AUC remain stable across different resolutions, especially on the LC25000 dataset, where AUC remains at 1.0 across all tested resolutions. This suggests that although downsampling leads to some loss of visual details, it does not significantly impact the model's performance in classification tasks.

Method	LC25000 Dataset				BreakHis			
	0.9* original size (691*691)	512*512	384*384	224*224	0.9* original size (630*630)	512*512	384*384	224*224
SSIM	0.9956	0.9895	0.9820	0.9308	0.9930	0.9894	0.9774	0.9261
PSNR	49.2252	45.4749	43.2162	37.2733	45.2979	43.5613	40.1482	34.8790

Table S12 Evaluation of image detail loss. All scores are averages

SSIM (Structural Similarity Index):

SSIM(Structural Similarity Index) [16,18] is a metric used to measure the similarity between two images, aiming to assess the visual impact of image quality changes. Unlike traditional metrics such as MSE (Mean Squared Error), SSIM considers human visual perception by comparing images in terms of three components: luminance, contrast, and structure.

PSNR (Peak Signal-to-Noise Ratio):

PSNR (Peak Signal-to-Noise Ratio)[17,19] is a metric that evaluates image quality based on the error between an original image and its processed version. PSNR is commonly used to quantify the loss of quality after compression or processing by measuring the ratio between the maximum possible pixel value and the noise (error) introduced.

SSIM and PSNR are the full-reference image quality assessment method. SSIM ≥ 0.98 or PSNR >> 40 indicates that the image is nearly indistinguishable from the original. 0.95 ≤ SSIM < 0.98 or PSNR ≥ 40 indicates acceptable image quality. 0.90 ≤ SSIM < 0.95 or 30 ≤ PSNR < 40 suggests noticeable degradation, while SSIM < 0.90 or 20 < PSNR< 30 indicates poor image quality.

5. References to CNN Models: The statement referring to "the latest reported CNN models and weakly supervised models" must specify which CNN models are being referenced. A lack of literature citations makes this point vague.

Thank you for your valuable comment. We have revised the manuscript to more clearly specify the referenced CNN, weakly supervised, and self-supervised models, along with appropriate literature citations. Specifically, we have added the following statement on Page 5, line 141 in the revised manuscript: "*BEPH achieved an average accuracy (ACC) of 94.05 ± 0.9928 and 93.65 ± 0.6730, respectively, which is 5% to 10% higher than the latest reported CNN models (Deep[31], SW[31], GLPB[38], RPDB[39]) , weakly supervised models (MIL-NP[32], MILCNN[32]) and self-supervised model (MPCS-RP[40]).*" This addition clarifies the models being referenced, ensuring clarity in the comparative results.

6. Figure Formatting: In Figure 2.f, the size of the caption is excessively large, which may distract from the content. In Figure 2.e, providing additional details about model size would be beneficial.

Thank you for your suggestion. We have reduced the font size of the caption in Figure 2.f. In Figure 2.e, we have added additional details regarding the model size based on either the reported structure in the corresponding studies or calculations from publicly available code. For models where neither source provided sufficient information, we have labeled them as "Details Unknown" to indicate the lack of available data on model size.

You can see our revisions in Figure 2 of the main manuscript or in Figure S2 (Page 6 in this reply), as they are identical.

7. Claim Regarding Clinical Potential: The assertion that “MIM-based pre-training enhances the adaptability of the model to pathological image resolution and cancer type transformation” lacks validation. The clinical potentialis not demonstrated, particularly given that the dataset used in this study is relatively small and derived from a single institution.

Thank you for your valuable feedback on our statement regarding the adaptability of MIM-based pre-training. The original manuscript is in page 5, line 150:

”*These results show that MIM-based pre-training enhances the adaptability of the model to pathological image resolution and cancer type transformation, and robustness to low-resolution image information, demonstrating the clinical potential of BEPH.*”

Our intention was to highlight that the BEPH model, through MIM-based pre-training, demonstrates consistent performance across multiple cancer types and resolutions within our dataset. This adaptability suggests that BEPH can be effectively applied to different cancer studies, rather than implying clinical application or broad clinical potential at this stage.

This ambiguity may have arisen from the phrasing in our manuscript, particularly regarding “cancer type transformation.” Consequently, we have revised the text (on page 5, line 153 in the revised manuscript) as follows:

" *These results demonstrate that MIM-based pre-training allows the BEPH model to effectively generalize across multiple cancer types, achieving strong performance on diverse pathological image data. This suggests that BEPH is robust to variations in image resolution and cancer type, making it potentially applicable to a wide range of cancer studies*."

8. Section Organization: The overall organization of the sections requires improvement for better clarity and flow.

We appreciate the reviewer's comment regarding the organization of the manuscript's sections, especially the **Methods** section. To enhance clarity and flow, we have restructured the manuscript as follows:

1. Section Reorganization: We have revised the order and content within key sections to improve logical progression and readability, ensuring that each section naturally builds upon the previous one.

2.Enhanced Transitions: We added transitional statements between sections to guide readers more smoothly through the manuscript and strengthen the coherence of the overall narrative. We believe these changes have improved the manuscript's structure and flow, and we hope this addresses the reviewer's concerns.

9. Misidentification of ResNet Models: The claim that "the BEPH outperforms the ResNet supervised pre-trained" lacks specificity; which ResNet model is referenced here?

Thank you for highlighting the need for specificity regarding the ResNet model referenced. We apologize for the oversight. In the revised manuscript, we have clarified that the ResNet model used for comparison is specifically ResNet-50. We have replaced all instances of "ResNet" with "ResNet-50" throughout the manuscript, where necessary (e.g. page 2, line 37:" *BEPH achieves up to 8.8% and 7.2% improvements over **ResNet-50** and DINOv1, respectively, in WSI-level classification, with an average improvement of 6.44% and 3.28% in survival prediction.*", and page 7, line 201:" *BEPH outperforms **ResNet-50** trained on ImageNet-1K in a supervised method (named to as ResNet-ImageNet)by 8.8%, DINO pre-trained on 256×256 histopathological images (named as DINO-HistoPretrain) by 7.2%.*"

10. Consistency in Formatting: There is inconsistency in formatting, such as "11,774,353 samples" versus "1.28 million."

Thank you for your feedback. We have reviewed the manuscript and standardized the formatting for large numbers to use the "million" format throughout. The necessary changes have been made in the revised version. You can see the changes on Page 2, Line 32:"

*leverages self-supervised learning to learn meaningful representations from **11 million** unlabeled histopathological images."* And page 15, line 361:*" We trained a histology-specific feature extractor on over 10,000 TCGA slides and approximately 11 million sampled unlabeled patches.*".

11. Comparison with Existing Datasets: The authors state, "Firstly, although our self-supervised pre-training dataset is much larger compared to currently reported datasets." This is misleading as referenced in the paper by Xu et al. [2], which reports "Prov-GigaPath, a whole-slide pathology foundation model pretrained on 1.3 billion 256 × 256 pathology image tiles in 171,189 whole slides from Providence, a large US health network comprising 28 cancer centers."

We appreciate the reviewer's insightful comment regarding the comparison of our dataset size to the recently reported large-scale datasets. When we initiated this work, there were relatively few studies in the field of pathology image analysis using large-scale self-supervised learning approaches, and at that time, our dataset was considered large compared to the commonly used datasets. However, we acknowledge that with the recent publications of datasets like Prov-GigaPath, which contains 1.3 billion pathology image tiles, our pre-training dataset is no longer among the largest in the field.

Nevertheless, we believe that our dataset, which consists of 11.77million samples, still meets the requirements for effective pre-training of the ViT-base model and compares favorably to many datasets used in prior studies. We also emphasize that our dataset provides sufficient diversity and volume to achieve robust performance in our specific task applications. We hope this clarifies the context in which our work was conducted and appreciate the reviewer's understanding of the evolution of available datasets in this rapidly progressing field.

Based on your feedback, we have revised these sections in the manuscript. The changes can be found from Page 14, Line 381 to Page 14, Line 387:

" Secondly, the pre-training dataset we have collected is consistent in scale with those used in many previously published studies, offering the necessary quantity and cancer

diversity to ensure model performance. However, the dataset size is smaller compared to those used in many currently published models, such as the Prov-GigaPath48, and it does not include multi-center data. While BEPH performed better on the survival prediction task, UNI and GigaPath performed better on the WSI classification task indicates that larger and more diverse datasets tend to better performance.".

12. Training Time Clarification: The authors mention that "Finally, our model is pretrained for 340 hours" without stating the stop-training criteria or the resources utilized.

Thank you for your valuable feedback. We have revised the manuscript in the **METHODS** section under **BEPH architecture and implementation** to clarify the training time. Specifically, we have provided details on how the training duration of 340 hours was determined based on the observed loss and previous studies. Additionally, we have included relevant details such as batch size, learning rate, and mask ratio used in the pretraining process. The changes can be found from page 17, line 454 to line 460: "*In our experiments, the model parameter size is 192.55 M. The image size is 224×224 and the batch size is 1024. The optimizer is AdamW and the initial learning rate is 0.0015. Finally, our model was pre-trained for a total of 340 hours using 8 NVIDIA A100 GPUs. The pre-training duration was based on observations of the pre-training loss, which gradually plateaued as training progressed. Additionally, the pre-training time was chosen in alignment with similar studies by Zhou et al[15], Bao et al.[27], and Wang et al.[10], where similar model architectures and pretraining procedures were followed.*"

"

13. Lack of Segmentation Details: The statement "WSI begins with the automatic segmentation of the tissue regions" lacks detailed methodology. More information is needed on how this segmentation was achieved.

Thank you for your comment. We have added detailed methodology regarding the automatic segmentation of tissue regions, including the use of the Otsu thresholding method to generate a binary mask and the steps involved in determining the optimal threshold for segmenting the foreground and background. The segmentation details can be found in section **"Processing WSIs",** from page 16, line 409 to page 16, line 425:" *Then, a binary Segmentation mask of the foreground and the background is generated by thresholding the saturation channel of the image. Specifically, we employed the histogram-based Otsu image segmentation approach. First, the input image is converted to grayscale, and its histogram is constructed by counting the number of pixels at each gray level (0 to 255). Then, based on a threshold* t*, the foreground and background regions are segmented. The optimal* t *is determined by maximizing the inter-class variance.*

$$\sigma_B^2(t) = w_1(t) \cdot w_2(t) \cdot (\mu_1(t) - \mu_2(t))^2$$

In the formula, $\sigma_B^2(t)$ *is inter-class variance,* $w_1(t)$ *and* $w_2(t)$ *are the pixel ratios of foreground and background.* $\mu_1(t)$ *and* $\mu_2(t)$ *are the average grayscale values of the foreground and background. Ultimately, we only retain the foreground regions with pixel values greater than the threshold* t*, which are tissue regions in the slide, typically. Besides, considering that some slides in the TCGA dataset may contain duplicated pathological tissue and the slide has a large size, it is unnecessary to include all the patches in pre-training. To preserve the contextual relationship between selected patches, we locally sample image regions of 1024 × 224 × 224 (approximately 1024 images) from each pathological image, ensuring that the sampled region has a tissue proportion greater than 75%.*"

14. Robustness of Patch-Level Classification Task: The dataset utilized for patch-level classification contains samples from a single institution; therefore, the robustness of the method is neither investigated nor demonstrated.

Thank you for your feedback regarding the robustness of the patch-level classification task. We apologize for any misunderstanding caused by our use of the term "robustness." In our manuscript, the phrase "robustness to low-resolution image information" specifically refers to the BEPH model's ability to maintain performance when working with lower resolution images. This description is intended to highlight the model's adaptability to varying image resolutions within the current dataset, rather than asserting its general robustness across datasets from different institutions.

As you pointed out, our dataset originates from a single institution which could have led to some ambiguity in our manuscript's phrasing. Therefore, we have revised the relevant section. On page 5, line 153, we now state: " *These results demonstrate that MIM-based pre-training allows the BEPH model to effectively generalize across multiple cancer types, achieving strong performance on diverse pathological image data. This suggests that BEPH is robust to variations in image magnifications (Supplementary Fig. 4) and cancer type, making it potentially applicable to a wide range of cancer studies.*"

15. Dataset Links: The dataset section should provide links to each dataset utilized in the study.

Thank you for the suggestion. We have updated the Dataset section to include direct links to each dataset used in this study, ensuring easy access for readers. Each dataset is now listed with a hyperlink and a brief description of its source. Please see Section Datasets used in the experiment for the modified content (Page 24 in the revised manuscript) .

16. Inconsistent Terminology: Phrases such as "Our public BRCA," "Our public NSCLC," and "Our public RCC" are inappropriate since these datasets were not provided by the authors.

Thank you for pointing out the inconsistency in our terminology. This issue only occurs in the **Datasets used in the experiment** section. We agree that the phrases "Our public BRCA," "Our public NSCLC," and "Our public RCC" (Page 20 in original manuscript) could misleadingly imply that these datasets were provided by the authors. To address this, we have revised the terminology throughout the manuscript to avoid any potential ambiguity. We now refer to these datasets simply as "The public TCGA-BRCA dataset," "The public TCGA-NSCLC dataset," and "The public TCGA-RCC dataset." (Page 25, Line 723 in revised manuscript).

17. Careless Presentation: Overall, the text appears to be prepared in a careless manner, requiring substantial revision to enhance clarity and professionalism.

We appreciate the reviewer's constructive feedback on the presentation of the manuscript and acknowledge the need for improvements to enhance clarity and professionalism. In response, we have implemented the following changes:

1. Revised the presentation of various sections, particularly the Methods section, to enhance coherence and readability.
2. Conducted thorough proofreading to address grammar, spelling, and punctuation errors, and refined the language to be more precise and formal, minimizing any ambiguity.
3. Rephrased complex sentences and removed unnecessary words, ensuring conciseness and alignment with recommended standards.
4. Standardized the formatting of figures, including fonts, titles, and captions, to improve the overall presentation.

We believe these revisions have significantly improved the manuscript and hope they satisfactorily address the reviewer's concerns.

Reference:

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Dear Reviewers:

Thank you for providing the opportunity to revise our manuscript titled " A foundation model for generalizable cancer diagnosis and survival prediction from histopathological images" (ID: NCOMMS-24-32882). We sincerely appreciate the reviewers' thoughtful and constructive comments, which were invaluable for improving the quality and clarity of our manuscript. These suggestions have not only enhanced our paper but also provided important insights to guide our research.

We have carefully addressed all the reviewers' comments point by point and made corresponding revisions to the manuscript. Detailed responses to each comment and the resulting changes are provided in the following pages. Representative results have been highlighted in the response letter, and additional tables and figures presenting the full results are included for clarity and completeness.

We hope that the revisions meet the expectations of the reviewers and improve the overall quality of the manuscript. Thank you again for your time and consideration, and we look forward to your feedback.

Response to Reviewer #2

The authors addressed most of the comments; however, not all explanations are accurate. For example, "For the survival prediction task, BEPH achieves the best performance across most cancer types, except for CCRCC, where UNI demonstrates superior results. This highlights BEPH's strong performance in survival prediction across various cancer types.". However, based on the included table, this statement is not entirely accurate. BEPH outperforms UNI significantly in only two cases: CRC and PRCC. Additionally, there is a lack of statistical analysis, which means we cannot determine whether the observed differences are statistically significant.

Thank you for your detailed and insightful comment. We acknowledge that our previous explanation had some shortcomings, particularly in the phrasing and the lack of statistical analysis, which may have caused misunderstandings.

The conclusion of the above sentence is that we ranked five foundation models and found that BEPH achieves either the best or second-best performance across almost all tested cancer types (Figure S2 in this reply; Supplementary Fig. 9 in the supplementary; Table S1 in this reply; Supplementary Table 13 in the supplementary). In contrast, UNI and CHIEF only outperform BEPH in specific cancer types:

a) UNI outperforms BEPH in CCRCC.
b) CHIEF outperforms BEPH in BRCA and STAD.
c) However, BEPH consistently ranks second in these cases and achieves the highest average rank overall.

This proves that BEPH exhibits excellent and consistent performance across different cancer types, even though its parameter size and training data are not as large as other foundation models.

On the other hand, our decision not to perform statistical tests was guided by the conventions in related studies, many of which do not include statistical tests for C-index values. For example, in several published works on survival prediction tasks[1–4], the C-index values are presented as performance metrics without accompanying statistical tests.

Nevertheless, we now perform additional statistical comparisons Using paired t-tests and Cohen's d analyses. The detailed results are provided in Table S2 in this reply (Supplementary Table 5 in the supplementary). Key findings are as follows:

a) BEPH significantly outperforms CtransPath across the six cancer types with $p < 0.05$ or Cohen's $d \geq 0.7$.

b) UNI significantly outperforms BEPH in CCRCC with $p < 0.05$ and Cohen's $d \geq 0.8$.

c) No statistically significant differences were found between BEPH and CHIEF in BRCA and STAD, although CHIEF performed slightly better in these cancer types. For CCRCC, CRC, and LUAD, BEPH achieved Cohen's $d \geq 0.6$.

d) In terms of C-index, BEPH outperforms GigaPath. Statistical tests show Cohen's $d > 0.5$ in all cases except CCRCC. Notably, Cohen's $d > 0.5$ for BRCA, CRC, STAD and $p < 0.05$ for BRCA.

These findings confirm that BEPH exhibits outstanding and stable performance across various cancer types, further supporting its effectiveness. We have updated the corresponding sections in the supplementary to reflect these findings (Page 7, line 229):

"*Compared to the latest foundation models, BEPH demonstrated excellent and consistent performance, achieving either the best or second-best rank in almost all cases (Supplementary Figure 8, Supplementary Table 13). Statistical analysis showed that BEPH significantly outperformed CtransPath across almost all cancer types ($p < 0.05$, Cohen's $d \geq 0.7$) and GigaPath in terms of C-index (Cohen's $d > 0.5$, except for CCRCC). Although UNI and CHIEF outperformed BEPH in specific cancer types (UNI in CCRCC; CHIEF in BRCA and STAD), BEPH maintained the second-best rank and the highest average rank, confirming its robustness and stable performance (Supplementary Table 5).*"

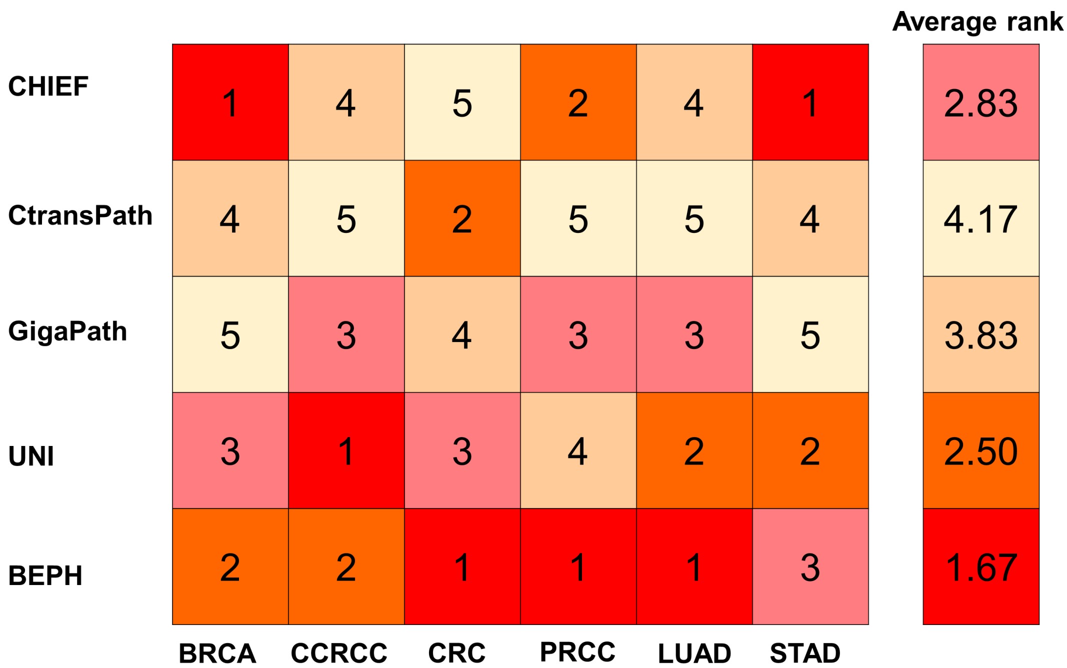

Figure S1 | In the survival prediction task, several foundation models are ranked by performance on six different cancers. The average rank is calculated by summing and averaging the rankings of the six cancers.

Method	BRCA	CCRCC	CRC	PRCC	LUAD	STAD
ABMIL	0.4871±0.0706	0.5612±0.0659	0.5656±0.0669	0.6712±0.0683	0.584±0.0481	0.5618±0.0439
DS-MIL	0.5338±0.0539	0.591±0.0834	0.5382±0.044	0.636±0.059	0.5918±0.0626	0.5131±0.062
GCN-MIL	0.4722±0.0183	0.5484±0.0511	0.4702±0.0477	0.6539±0.1195	0.5373±0.0543	0.5459±0.042
DeepAttnMISL	0.4716±0.0207	0.5207±0.0749	0.5613±0.0786	0.4718±0.1448	0.563±0.0327	0.5633±0.0596
HIPT	0.6344±0.0449	0.6416±0.0251	0.6081±0.0784	0.6698±0.0584	0.5377±0.0397	0.57±0.072
CLAMSurvival(ResNet)	0.5946±0.0484	0.6466±0.0405	0.5502±0.1187	0.599±0.1368	0.552±0.1002	0.5119±0.0882
CLAMSurvival(DINO)	0.5987±0.0316	0.6555±0.0807	0.6207±0.0399	0.6974±0.1105	0.5687±0.0549	0.584±0.0428
CLAMSurvival (CHIEF)	**0.686±0.0364**	0.6482±0.0643	0.4844±0.0635	0.6973±0.1518	0.5533±0.0941	**0.6372±0.0391**
CLAMSurvival(CtransPath)	0.5653±0.0473	0.6158±0.0251	0.6027±0.0658	0.559±0.1432	0.5116±0.0497	0.541±0.0378
CLAMSurvival (UNI)	0.6519±0.0652	**0.709±0.037**	0.6007±0.0866	0.6212±0.0703	0.5713±0.0909	0.6254±0.0248
CLAMSurvival(GigaPath)	0.5642±0.0279	0.6644±0.0251	0.5737±0.1266	0.6555±0.1182	0.5623±0.0521	0.5375±0.0571
CLAMSurvival (BEPH)	0.6643±0.033	0.6685±0.0386	**0.6758±0.0721**	**0.7135±0.1557**	**0.6039±0.0762**	0.5941±0.0478

Table S1. C-index across six cancer types, derived from five-fold cross-validation.

All data are presented as mean values +/- SD. Bold indicates the best results.

validation	BRCA		CCRCC		CRC		LUAD		PRCC	STAD		
	p-value	Cohen's d	p-value	Cohen's d	p-value	Cohen's d	p-value	Cohen's d	p-value	Cohen's d	p-value	Cohen's d
BEPH --ABMIL	0.0205	1.6637	0.1001	0.953	0.1292	0.8527	0.3623	0.1747	0.6891	0.1925	0.2065	0.6736
BEPH --GCN-MIL	0.0012	3.6999	0.2538	1.3216	0.0131	1.8736	0.8438	0.623	0.3917	0.278	0.047	0.6743
BEPH --DS-MIL	0.0033	2.7934	0.0417	0.5955	0.0138	1.9031	0.236	0.094	0.5678	0.429	0.2061	1.2693
BEPH --HIPT	0.3074	0.5227	0.0167	0.4675	0.0922	0.5525	0.2523	0.5572	0.1338	0.1888	0.5946	0.3781
BEPH--DeepAttnMISL	0.0002	5.6736	0.3548	1.7700	0.2843	0.9855	0.2808	0.5977	0.6946	0.8391	0.4456	0.2582
BEPH--CLAMSurvival(Resnet)	0.0137	1.8769	0.0245	1.572	0.0688	1.1053	0.1183	0.887	0.3837	0.4371	0.1192	0.8843
BEPH –CLAMSurvival(DINO)	0.0732	1.0796	0.6299	0.233	0.1225	0.8734	0.0916	0.9886	0.8179	0.1099	0.6854	0.195
BEPH –CLAMSurvival(UNI)	0.7889	0.1280	0.0357	1.3931	0.0529	1.2171	0.0574	1.1818	0.1878	0.7096	0.3773	0.4436
BEPH –CLAMSurvival(CHIEF)	0.5438	0.2964	0.2573	0.5904	0.0202	1.6718	0.0916	0.9884	0.6152	0.2434	0.2730	0.5679
BEPH –CLAMSurvival(CtransPath)	0.0115	1.9762	0.0489	1.2513	0.1406	0.8200	0.0171	1.7589	0.0023	3.0928	0.1744	0.7375
BEPH –CLAMSurvival(GigaPath)	0.0007	4.1753	0.8205	0.1084	0.0939	0.9782	0.1565	0.7790	0.2286	0.6351	0.1022	0.9446

Table S2. Two-sided paired t-test and Cohen's d effect size results. Bold indicates BEPH comparisons with other methods where p-value < 0.05 or Cohen's d > 0.8 (Small effect size when 0.2<Cohen's d<0.5, Medium effect size when 0.5≤Cohen's d<0.8, Large effect size when Cohen's d≥0.8), demonstrating significant differences between methods.

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